

INVESTMENT REQUIRED

\$3.0 Million

ANNUAL SAVINGS BY 2028

\$20.5 Million

NET BENEFIT

\$17.5 Million

216,100 human hours/year eliminated • 23 platforms in ROI scope • ~6.8x return on investment

THE OPPORTUNITY

The Zephyr On-Prem IaC initiative automates 34 manual and semi-automated operational tasks across five efficiency categories: Onboarding, Provisioning, Management, Compliance, and Optimization. Across 23 platforms, this eliminates 216,100 hours of manual effort annually.

HOW IT WORKS

Net New Asset Provisioning: 275 mins/asset × 20 assets/yr × 23 platforms = **2,100 hrs/yr**

Existing Asset Maintenance: 186 mins/asset/yr × 3K assets × 23 platforms = **214,000 hrs/yr**

Total Hours Eliminated: 2,100 + 214,000 = **216,100 hrs/yr**

Financial Impact: 216.1K hrs × \$95/hr = **\$20.5MM/yr**

Net Benefit: \$20.5MM – \$3.0MM investment = **\$17.5MM**

MIGRATION WORKSTREAM

A dedicated migration workstream in EPS partnering with ZOP is required to transition legacy platforms to IaC. This includes configuration standardization, asset migration into the IaC management plane, legacy process retirement, and team training. Without this workstream, the full \$17.5MM net benefit cannot be realized.

PLATFORM SCOPE (23 OF 34 IN ROI)

Category	Count	Key Services
Application Hosting	9 platforms	Websphere, JBOSS, PEGA, ISAM, ESM, Message Broker
Database	4 platforms	MongoDB, SCCN, SQL Server, MemSQL
Networked Storage	4 platforms	NAS, File Hosting, Backup & Recovery, SAN
Compute & Containers	1 platform	Diamanti
Network Services	4 platforms	Extranet, DNS, CDN, Datacenter Networking
Mainframe	1 platform	Messaging (z/OS)

Phased rollout: Q1 2026 – Early 2027

KEY ASSUMPTIONS

- 20 net new assets/platform/year (conservative avg)
- 275 mins per net new asset provisioning
- 3,000 existing assets per platform
- 186 mins/yr per existing asset maintenance
- Avg FTE compensation: \$95/hr
- Estimates are conservative — DB volume (~20 req/week) and asset counts (SQL Server ~1,500 instances alone) suggest material upside

STRATEGIC BENEFITS

- Eliminates configuration drift and human error
- Accelerates time-to-deploy for new infrastructure
- Version-controlled audit trail for compliance
- Scalable to remaining 11 platforms for added ROI

RECOMMENDATION: Approve the \$3.0MM investment to proceed with phased IaC rollout across 23 platforms (2026–2027). The dedicated EPS + ZOP migration workstream will deliver \$17.5MM net benefit — a ~6.8x return — with conservative assumptions suggesting material upside.

The \$3.0MM investment and \$17.5MM net benefit assume traditional engineering delivery. Layering an AI-SDLC “Codify Factory” on top of Zephyr On-Prem can improve unit economics by compressing legacy platform codification timelines and reducing per-platform engineering effort. Below are realistic estimates for an enterprise environment with regulated change management, multi-team coordination, and legacy platform complexity.

CODIFICATION TIME 40–50% Faster Per-platform IaC authoring reduced, though review cycles and change governance remain	ENGINEERING EFFORT 30–40% Reduction AI agents draft IaC, API specs, and playbooks; humans review, adjust, and approve	FULL ROLLOUT TIMELINE 12–18 Months Compressed from 18–24 months; approval gates and platform complexity bound the floor	QUALITY COMPOUND Incremental Gains Factory learns patterns over time; later platforms benefit from earlier codification
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THE AI-SDLC “CODIFY FACTORY” PIPELINE

1. Discover & Harvest AI agents ingest legacy runbooks, SOPs, gold images, vendor best practices, and live config state from existing environments. Output: normalized knowledge base per platform.

2. Author IaC & Playbooks Agents draft IaC modules, Ansible roles, Zephyrize platform API specs, and hardening. Agents propose; engineers review and deploy the code further as per RBC standards.

3. Generate Tests Automated generation of test suites and policy-as-code checks. AI produces both happy-path and edge-case coverage aligned to platform standards.

4. Human Review Gate Engineers validate all agent-authored artifacts. “AI proposes, human decides” — nothing reaches production without human approval and change governance.

5. Publish to Registry & Catalog Validated modules published to the IaC registry and ServiceNow catalog. Hardened-by-default, self-service-ready offerings.

6. Operate & Feedback Drift detection, compliance monitoring, and incident signals feed back into the factory, improving future codification cycles.

WHAT AI-SDLC CANNOT AUTOMATE

- Access approvals and entitlement workflows — require human authorization per RBC security policy and regulatory guidelines
- Change Advisory Board (CAB) reviews and production change governance — regulatory mandates require human accountability
- Manual decision-making on architecture, placement, and capacity — context-dependent judgment that agents cannot replace
- Vendor-specific platform limitations — legacy APIs, proprietary tooling, and vendor lock-in that constrain automation scope
- Cross-team coordination and stakeholder alignment — organizational dependencies that exist outside the technical pipeline
- Regulatory interpretation and risk acceptance — compliance decisions requiring human sign-off per enterprise risk and regulatory frameworks

UNIT ECONOMICS: MANUAL vs. AI-SDLC

Metric	Manual	AI-SDLC
Per-platform codification	14–16 weeks	6–8 weeks
Engineer FTEs per platform	2–3 FTEs	1–2 FTEs
Test coverage generation	Manual, partial	AI-assisted, ~60–70%
Hardening validation	Checklist-based	Policy-as-code
Time to catalog-ready	6–8 weeks	3–5 weeks
Drift detection setup	Custom scripting	Agent-assisted config
23-platform full rollout	18–24 months	12–18 months

BUSINESS CASE IMPACT

Faster Time-to-Value: Full rollout in 12–18 months vs. 18–24 means savings begin **6–12 months earlier**. Internal process gates (CAB, access, vendor coordination) remain the binding constraint.

Reduced Delivery Cost: 30–40% less engineering effort per platform frees capacity within the \$3.0MM investment for additional platforms or deeper hardening coverage.

Compliance Acceleration: Auto-generated policy tests and audit lineage reduce mean-time-to-evidence for compliance and regulatory audits. Approval authority remains human.

Scalability Beyond 23: The factory makes onboarding the remaining 11 non-ROI platforms more economically viable, unlocking incremental savings at lower marginal cost.

Governance Preserved: AI-SDLC accelerates the technical work; it does not bypass change governance, access controls, or regulatory gates. Human accountability remains at every decision point.

AI-SDLC AS ACCELERATOR, NOT REPLACEMENT: The Codify Factory compresses what can be compressed — IaC authoring, test generation, and configuration harvesting — while respecting what cannot: change governance, access approvals, regulatory sign-off, and human decision-making. The result is a 6–12 month acceleration in realizing the \$17.5MM net benefit, with full governance and accountability preserved.